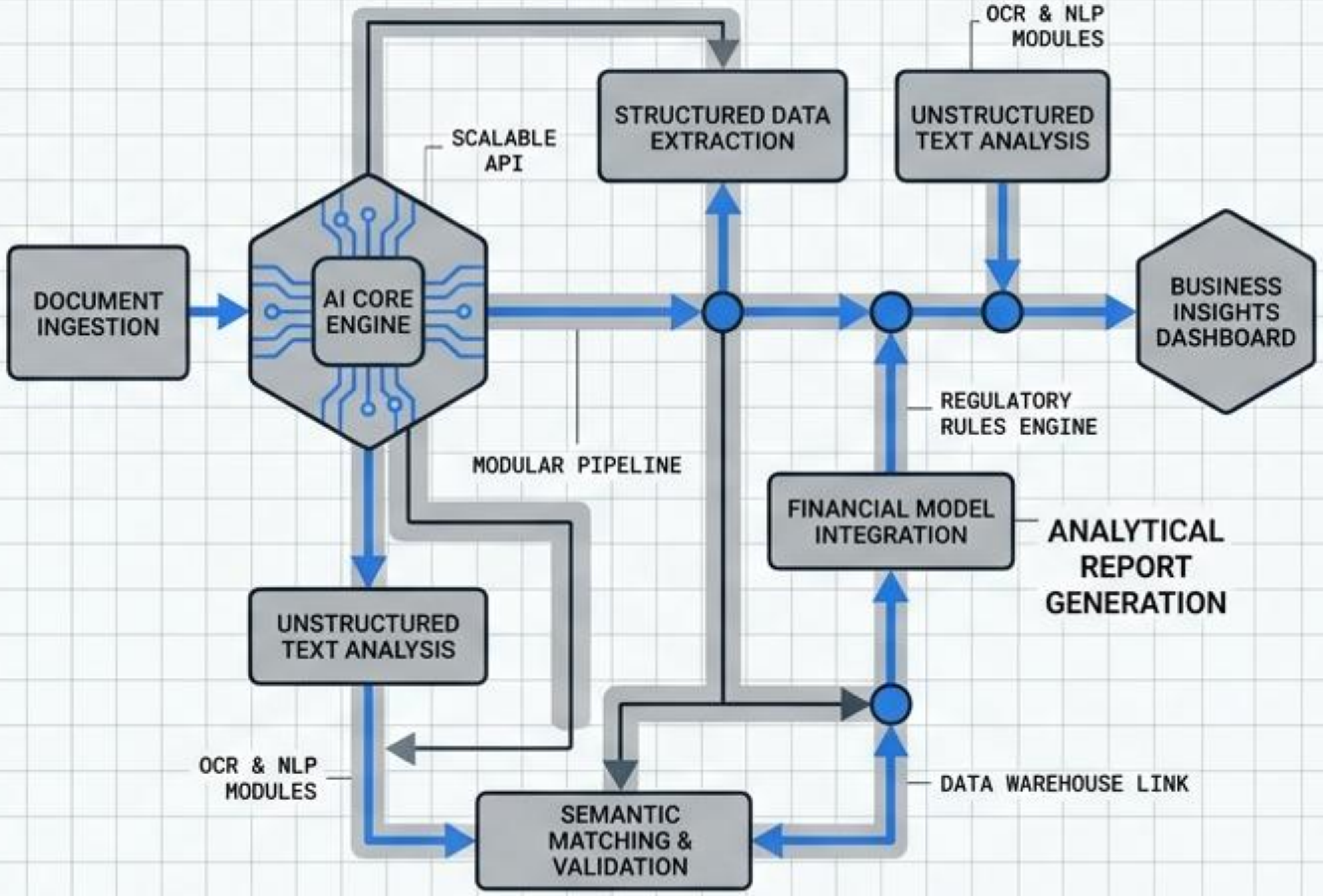
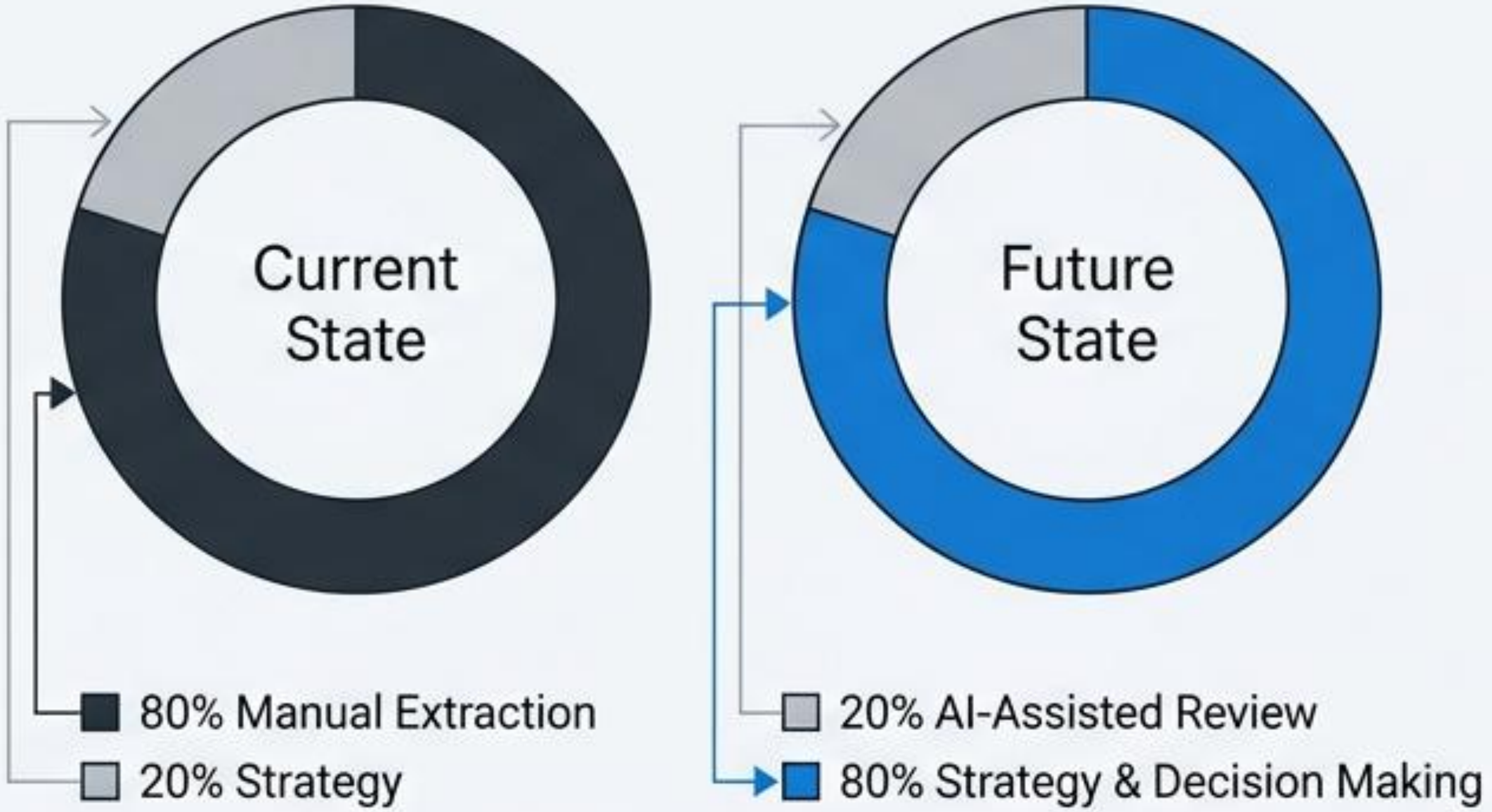
Architecting Reliable AI for Financial Statement Analysis

An MVP Case Study in Problem Scoping, Modular System Design, and Business Impact.

Production AI is not built on one powerful prompt. It is built on resilient architecture.



THE CRITERIA FOR AUTOMATION



- Is the workflow highly repetitive?
- Is the data structured enough for AI automation?
- Does automating it create measurable business value?

The goal is to augment, not replace. Free up human bandwidth for high-leverage investment decisions.

System Design: Monolithic vs. Modular

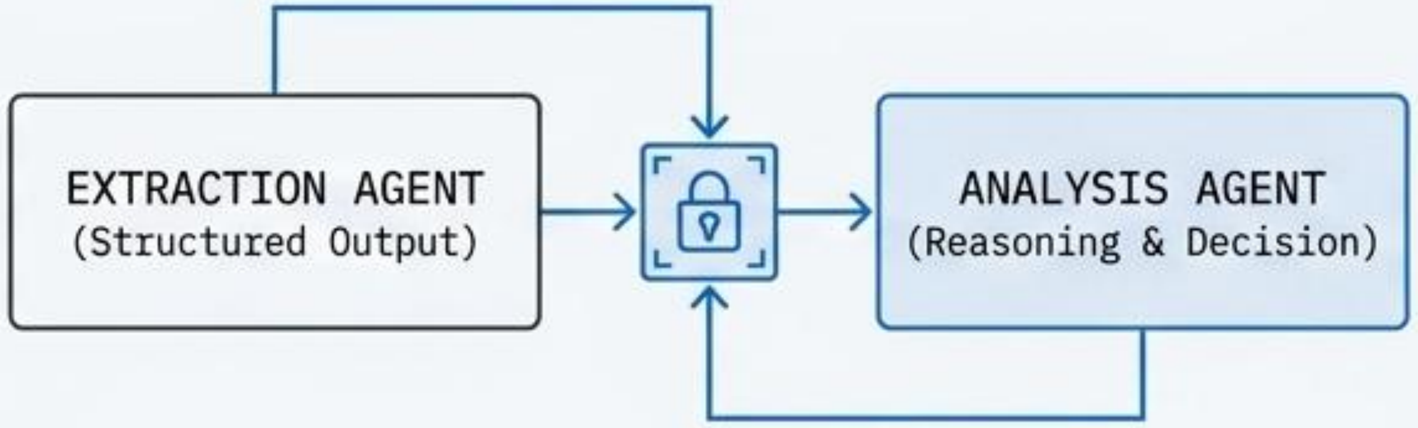
The Anti-Pattern

The God Prompt (Single Model)

1	**Reliability:** High hallucination risk.
2	**Maintainability:** Black box debugging.
3	**Scalability:** Brittle to edge cases.

The Engineered Approach

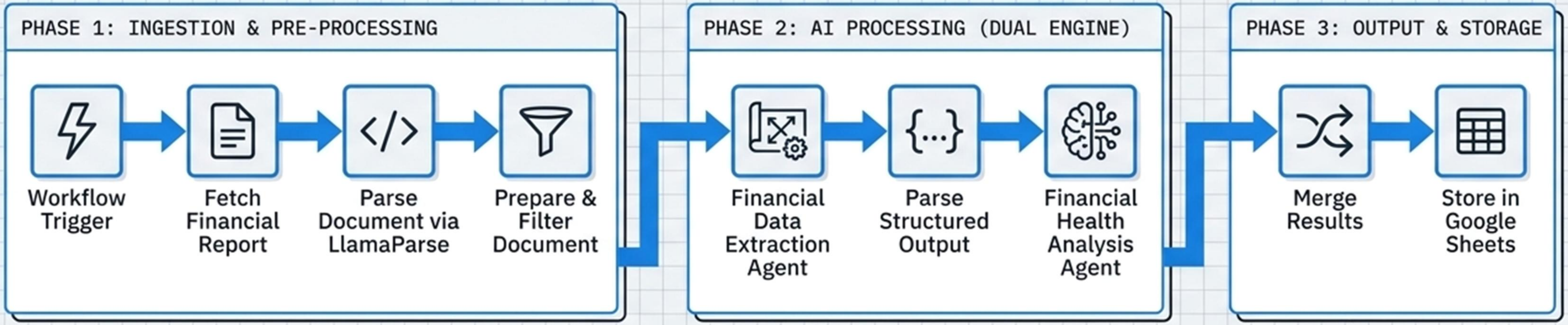
Decomposed Multi-Agent Pipeline



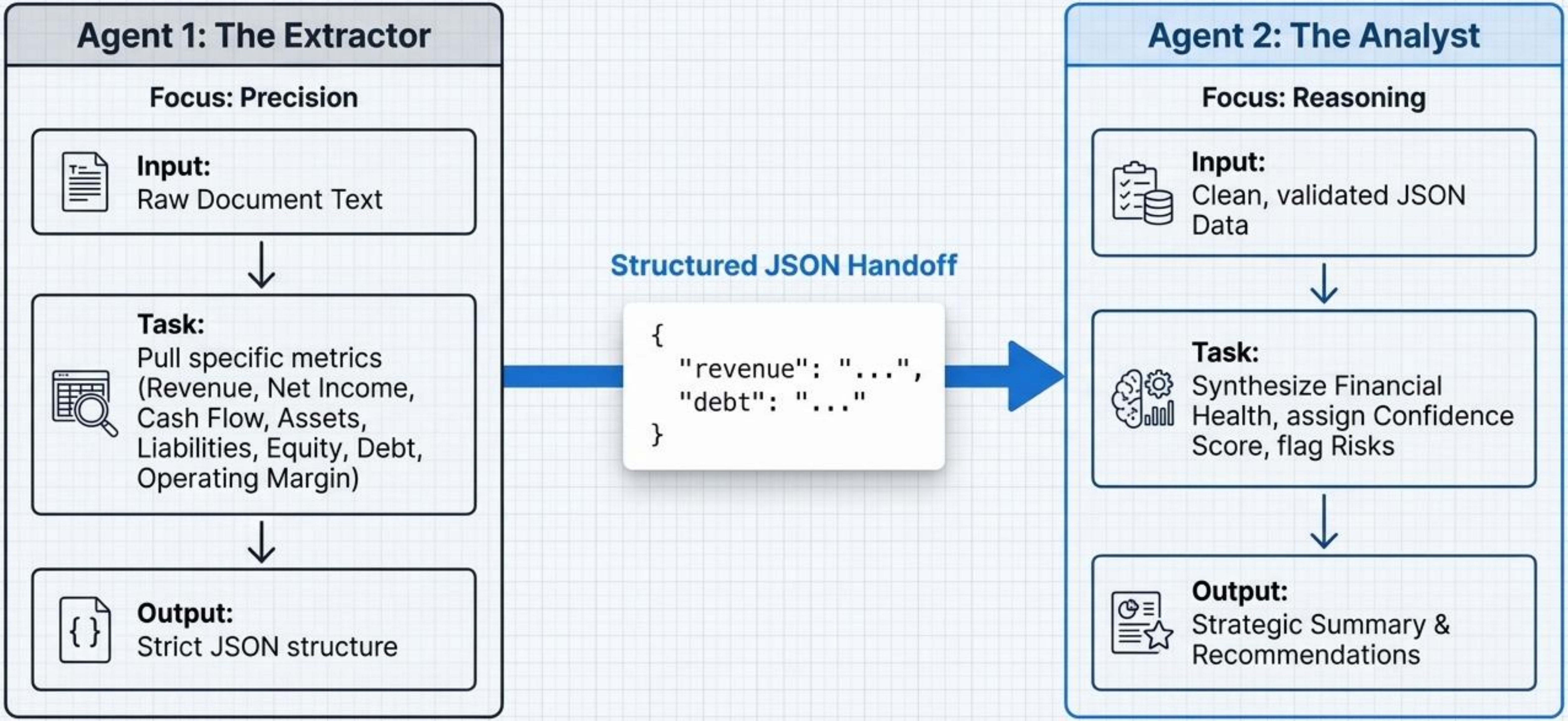
1	**Reliability:** High precision and strict validation.
2	**Maintainability:** Easily debuggable isolated steps.
3	**Scalability:** Independent component upgrades.

Architectural Choice: Separate the computational problem of Extraction from the reasoning problem of Analysis.

The End-to-End Pipeline Architecture



Separation of Concerns: Extraction vs. Analysis



Continuous Iteration: Building for the Real World

Challenge		Fix
Terminology: Different terminology across industries (Banks vs. Manufacturing)	→	**Fix:** Implemented industry-specific extraction rules
Structure: Investor presentations containing multiple companies or parent/subsidiary confusion	→	**Fix:** Added parent company validation layers
Scale: Varying reporting units (Crores vs. Rupees/Millions)	→	**Fix:** Built automated unit normalization logic
Stability: General LLM output instability and formatting errors	→	**Fix:** Enforced strict JSON validation and structured parsing prior to analysis

The Path to Production

Step 1: MVP Limitations (Current State)

- Token limit exhaustion on massive annual reports.
- Complex nested table extraction vulnerabilities (OCR limits).
- Manual handling of specific balance-sheet debt parsing.

Step 2: V2 Roadmap (Prioritizing Reliability)

Infrastructure: Implement a Vector DB to retrieve only relevant financial statement pages.

Accuracy: Integrate table-aware extraction models.

Safety: Establish Confidence-based Human-in-the-Loop (HITL) review routing.

UX/Integration: Transition from Google Sheets to Custom Dashboard and internal API.

AI as an Output Multiplier

The ROI

Faster Cycles

Reduced manual data entry leads to exponentially higher Analyst ROI.

Data Moat

Creation of a proprietary, reusable, structured financial database.

Strategic Focus

Human remains in control of investment decisions; AI owns the data processing.

The Bottom Line

> Great AI products aren't built on prompts. They are built on structured problem decomposition, resilient architecture, and human-centric design.